

Lab Experiment-VI: Reynolds Apparatus

Objectives

- (i) Observe transition between laminar and turbulent flows in a circular tube.
- (ii) Experimentally estimate the critical Reynolds number prior to transition.
- (iii) Observe the behavior of a passive tracer in laminar and turbulent flow regimes.

The Essential Background

- 1. How is the Reynolds number (Re) defined for a flow? How is Re defined for flow through a circular tube?
- 2. What is the physical significance of the Reynolds number? Show that it signifies the ratio between inertial and viscous forces in a fluid flow.
- 3. How do you define a laminar and a turbulent flow? What causes a laminar flow transition into a turbulent one?
- 4. What is the critical Re (Re_{cr}) for flow through a pipe, when the transition takes place? Do all flows have the same Re_{cr} at transition? If so, why? If not, report the Re_{cr} for various types of flows (such as flow over flat plates, flows around cylinders, etc.).

Task 6.1: In the report for Lab Exp.-VI , concise answers to all the above questions have to be submitted.

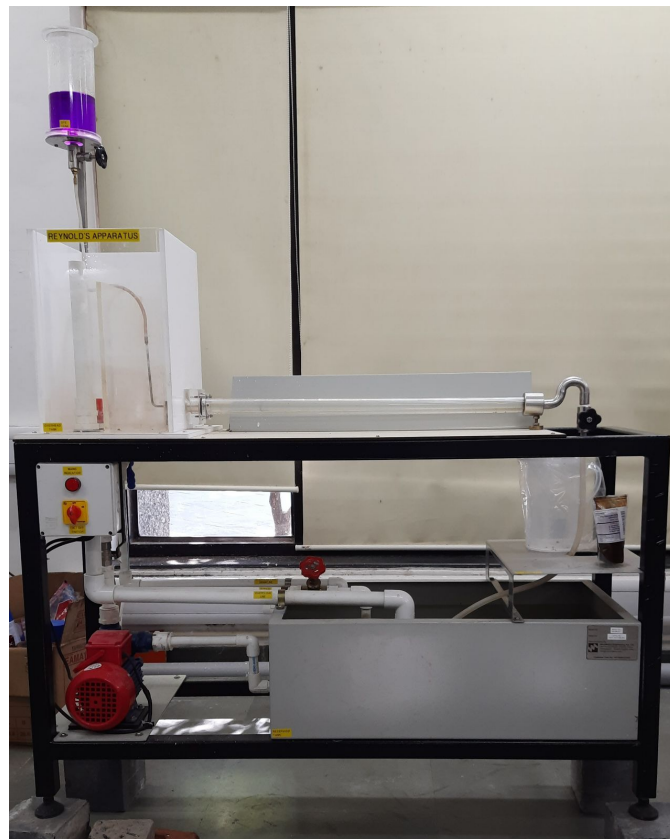


Figure 6: Set up of the Reynolds apparatus.

About the experiment

The experimental set up for the Reynolds experiment has been shown in figure 6. This is a simple set up where liquid water is flown through a circular tube. The water is pumped from a sump tank to a reservoir and from there it flows to the tube. To visualize the flow field, a colored die (soluble in water) is introduced at the inlet of the tube from a separate reservoir. The die will follow a straight line if the flow in the tube is laminar. On the other hand, it will fluctuate stochastically upon transition.

Task 6.2:

- (a) Start the flow in the tube at its minimum. Introduce the die in the tube by opening the appropriate valve. Observe the die particles following more-or-less a straight line inside the channel.
- (b) Now, slowly increase the flow rate by controlling the valve attached with the main supply line. Observe the instant at which the flow transitions into being turbulent by noting the behaviour of the die.
- (c) Measure the flow rate and evaluate the Re_{cr} . Repeat this experiment multiple times and check whether you get the same Re_{cr} each time.
- (d) Report experimental uncertainties and quantify errors.

Tasks 6.1 and 6.2 should be included in the report for this experiment.